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Hyperparameter optimization of graph neural networks for predicting complex network dynamics using bayesian meta-learning

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ABSTRACT

The rapid growth of graph-structured data in domains such as transportation, social networks, and biological systems has increased the demand for more adaptive and efficient Graph Neural Network (GNN) architectures. However, GNN performance remains highly sensitive to hyperparameter configurations, which are often tuned through computationally expensive manual or heuristic methods. This study proposes a novel Bayesian Meta-Learning (BML)-based framework for hyperparameter optimization of GNNs aimed at improving the prediction accuracy of complex network dynamics. The framework integrates Bayesian optimization with a meta-learning prior adaptation mechanism, enabling the model to learn optimal hyperparameter distributions across multiple graph tasks. Experimental evaluations conducted on three benchmark datasets-Cora, Citeseer, and PubMed-comprising up to 20,000 nodes with diverse structural complexities, demonstrate that the proposed BML-GNN framework achieves faster convergence, lower validation loss, and higher predictive accuracy than both baseline GNN and traditional Bayesian Optimization approaches. Quantitatively, the BML-GNN model attains an R^2 score exceeding 0.97 with a significant reduction in RMSE, confirming its strong generalization capability. Although the method shows notable performance improvements, its computational overhead during meta-training and reliance on well-defined prior distributions represent potential limitations. Overall, the integration of Bayesian Meta-Learning provides a robust, scalable, and uncertainty-aware optimization strategy that advances the development of reliable GNN models for complex network modeling and intelligent system design.

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1. INTRODUCTION

The rapid development of artificial intelligence (AI) and machine learning (ML) technologies has driven improvements in the capabilities of predictive systems to analyze complex phenomena that represent relationships between entities (Oladele Junior Adeyeye & Ibrahim Akanbi, 2024; Rane et al., 2024). One of the most widely used forms of representation for modeling such systems is graph-based data representation, where nodes represent entities and edges represent relationships or interactions between them. Graph structures are found in various domains, such as social networks, recommendation systems, transportation, communication, molecular biology, and power grids (Manurung & Saragih, 2024; Ponzi & Napoli, 2025; Sharma et al., 2024). In this context, the need for

predictive models capable of understanding complex network dynamics has become increasingly important to support accurate and efficient data-driven decision making (Manurung et al., 2025; Manurung & Sihotang, 2025). Graph Neural Networks (GNN) have emerged as one of the most effective approaches for learning patterns and relationships in graph-based data (Asif et al., 2021; Waikhom & Patgiri, 2023). GNN combines the principles of representation learning with message passing mechanisms, enabling the model to extract information from the topological structure and node features simultaneously. This approach has successfully improved performance in various prediction tasks, such as node classification, link prediction, and graph-level regression. However, GNN performance is highly influenced by hyperparameter configuration-such as layer depth, hidden dimension, learning rate, activation function, and inter-node aggregation strategy. Selecting the optimal combination of hyperparameters is crucial, as errors in tuning can result in underfitting, overfitting, or slow convergence (Yang et al., 2024; J. Zhou et al., 2023).

Most previous studies still rely on conventional exploration approaches for hyperparameter optimization, such as grid search, random search, or manual tuning (Ali et al., 2023; Budihal et al., 2023; Iqbal et al., 2018). These methods have major drawbacks, namely high computational requirements and inefficiency in large search spaces. In the context of GNNs, which have complex model structures, hyperparameter space exploration becomes very costly, both in terms of time and computational resources. To overcome these limitations, the Bayesian Optimization (BO) approach has been widely used as a more efficient alternative. BO works by building a probabilistic model (usually a Gaussian Process) to estimate the objective function and select the most promising combination of hyperparameters based on a balance between exploration and exploitation. However, even though Bayesian Optimization offers significant efficiency compared to conventional methods, this approach still has limitations in the context of cross-task adaptation. Every time it is applied to a new dataset or domain, the optimization process needs to be repeated from scratch without utilizing previous optimization experience. In fact, many prediction problems in complex networks have similar data structures or distributions. Thus, a mechanism capable of “learning from previous optimization experiences” would be very useful in accelerating the process of finding optimal hyperparameters. One promising approach to address this challenge is meta-learning, often referred to as learning to learn. In the context of model optimization, meta-learning aims to learn general patterns from a number of previous optimization tasks, then transfer that knowledge to new tasks so that the hyperparameter tuning process can take place more quickly and efficiently. The integration of Bayesian Optimization and Meta-Learning-known as Bayesian Meta-Learning-enables the construction of adaptive optimization models that are not only computationally efficient, but also capable of cross-domain generalization. This approach has shown promising results in other fields such as convolutional neural network (CNN) optimization and time series models (LSTM), but its application to GNNs is still relatively limited and has not been explored in depth (Ahmed et al., 2025; Corradini et al., 2025).

Several recent studies have focused on improving the efficiency of hyperparameter optimization for GNNs through heuristic approaches, such as genetic algorithms, particle swarm optimization, and reinforcement learning (Yuan et al., 2021). However, these methods often struggle to strike a balance between exploration and exploitation, and lack strong knowledge transfer capabilities between tasks. On the other hand, research combining Bayesian Optimization with GNN is still limited to a single optimization process, without a meta-learning mechanism capable of accelerating adaptation to new datasets. Therefore, there is a significant research gap, namely the absence of a Bayesian Meta-Learning-based hyperparameter optimization framework specifically designed for GNN in the context of predicting complex network dynamics. Considering this gap, this study aims to develop and implement a Bayesian Meta-Learning-based GNN hyperparameter optimization framework. The proposed approach will utilize the capabilities of Bayesian Optimization in efficiently exploring the hyperparameter space and the capabilities of meta-learning in transferring information from previous optimization experiences. Through this approach, it is expected that the model can accelerate the process of searching for the optimal hyperparameter combination, improve prediction accuracy, and significantly reduce computational costs. Conceptually, this research is expected to provide two main contributions. First, a theoretical contribution in the form of the development of a Bayesian Meta-Learning-based adaptive optimization model that can be applied to various GNN architectures, including Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), and Graph Isomorphism Networks (GIN). Second, a practical contribution in the form of improved prediction performance and training efficiency in real-world applications, such as

predicting information propagation on social media, power grid stability, and citation pattern analysis in academic networks. Thus, this research has the potential to expand our understanding of adaptive learning mechanisms in graph-based prediction model optimization and provide a methodological basis for the development of smarter, more efficient, and more generalizable predictive systems (Maldini, 2024; Munikoti et al., 2024).

In summary, the urgency of this research lies in the need for optimization models that are capable of learning from experience and adapting across domains. While previous research has focused on static optimization and been limited to a single domain, the proposed approach offers a new paradigm through the integration of Bayesian Meta-Learning to solve hyperparameter optimization problems dynamically and continuously. Through this approach, the process of developing complex network prediction models is expected to become faster, more efficient, and more robust against diverse data variations. Based on this background, the main objectives of this research are to:Design a Bayesian Meta-Learning-based hyperparameter optimization framework for Graph Neural Networks; Evaluate the performance of the proposed model in predicting complex network dynamics by comparing it to conventional optimization methods; and Assess the extent to which this approach can improve computational efficiency and the cross-domain generalization capabilities of prediction models (Vincent et al., 2025; Y. Zhou, 2025).

Graph Neural Networks (GNN) are one of the major innovations in machine learning, specifically designed to handle graph-based data, where relationships between entities are an important element in data representation. Unlike conventional models such as CNN or RNN that work on grid-structured or sequential data, GNN is capable of learning representations directly on non-Euclidean structures such as social networks, genetic relationships, or transportation systems (Jing, 2023). The core of GNN lies in the message passing mechanism, which is a process where each node updates its representation by aggregating information from neighboring nodes. This process allows the model to capture local and global dependency patterns within the graph. The most common GNN architecture variants include Graph Convolutional Networks (GCN) (Sarkar et al., 2023), Graph Attention Networks (GAT) (Vrahatis et al., 2024), and Graph Isomorphism Networks (GIN) (Liu & Wang, 2021). Each architecture offers specific advantages in terms of computational efficiency, representation capacity, and robustness to changes in network topology. In the context of predicting complex network dynamics, GNNs play an important role due to their ability to model changes in relationship structures over time. For example, in social networks, GNNs are used to predict patterns of information dissemination or user behavior. In power grids, GNNs help estimate system stability based on dynamic interactions between nodes. Although GNNs have demonstrated superior performance in various domains, their performance is highly dependent on the selection of appropriate hyperparameter configurations (Chen et al., 2024). Therefore, careful and efficient optimization strategies are needed to achieve optimal prediction results.

Hyperparameter optimization is an important stage in the development of machine learning models, including GNNs. Hyperparameters such as learning rate, number of hidden layers, batch size, activation function, and dropout rate greatly affect the generalization ability of the model. The traditional optimization process is usually done through grid search or random search, where parameter combinations are tried systematically or randomly (Bergstra & Bengio, 2012). Both methods are simple but computationally expensive, especially for complex models such as GNNs, which have a large search space and non-linear dependencies between parameters. As an alternative, several heuristic approaches such as Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Simulated Annealing (SA) have been proposed to accelerate the search process (Gad, 2022). However, heuristic methods often face challenges in terms of result stability, slow convergence, and inability to adapt to new domains without retraining from scratch. With increasing model complexity, there is a need for optimization approaches that are not only efficient but also adaptive to changes in data context. This has led to the application of probabilistic optimization approaches such as Bayesian Optimization (BO).

Bayesian Optimization (BO) is a probabilistic method used to find the optimal hyperparameter configuration with a minimal number of model evaluations. BO works by building a surrogate model, typically using a Gaussian Process (GP) or Tree-Structured Parzen Estimator (TPE), to model an objective function that is expensive to compute (Tahkola, 2019). Based on these posterior estimates, BO selects new search points through acquisition functions such as Expected Improvement (EI) or Upper Confidence Bound (UCB). The main advantage of BO is its ability to balance exploration and exploitation, thereby accelerating convergence towards the optimal solution

without the need to explore the entire parameter space. The application of BO to GNNs has shown significant performance improvements in several studies. For example, Hasanzadeh, Arman, et al. (Hasanzadeh et al., 2020) used Bayesian Graph Neural Networks to improve the efficiency of GNN architectures with an adaptive BO approach. However, BO has limitations in terms of knowledge transfer. When applied to new datasets, BO must build a surrogate model from scratch, without utilizing information from previous optimizations.

Meta-learning, or learning to learn, is an approach that enables models to learn how to learn more efficiently based on past experiences. In the context of optimization, meta-learning can be used to learn the distribution of previous optimization tasks and transfer these patterns or priors to new tasks (Finn et al., 2017). There are two main paradigms in meta-learning: (1) Model-based meta-learning, which uses neural models to predict parameter updates, and (2) Optimization-based meta-learning, which explicitly learns parameter update strategies, such as in Model-Agnostic Meta-Learning (MAML). The integration of meta-learning with Bayesian Optimization results in what is known as Bayesian Meta-Learning (BML)-an approach in which probabilistic models not only estimate the objective function but also learn from the distribution of previous optimization experiences (Ravi & Larochelle, 2018). In the context of GNNs, the application of BML is still relatively new. Several early studies (e.g., Chen et al., 2023) show that this approach can significantly accelerate hyperparameter search by leveraging meta-priors from similar datasets. This demonstrates the great potential of BML for use in domains where efficiency and adaptability are critical, such as in the prediction of complex network dynamics that have similar structures and distributions.

Previous studies have examined GNN optimization efforts through different approaches. For example, Li et al. (2021) proposed AutoGNN, which automates architecture and hyperparameter tuning using neural architecture search (NAS) techniques. Meanwhile, Zhou et al. (2022) applied Bayesian Optimization to tune the hyperparameters of GCN and GAT, and succeeded in increasing accuracy by up to 12% compared to grid search. On the other hand, meta-learning has been applied in the CNN and RNN domains to accelerate the training process, but its application in GNN is still limited to few-shot node classification settings (Huang et al., 2023). The main gap in the existing literature is the lack of an integrative approach that combines the probabilistic efficiency of Bayesian Optimization with the adaptive capabilities of Meta-Learning for hyperparameter optimization in GNNs. Most studies focus on only one aspect-efficiency or adaptivity-but have not combined the two in a comprehensive framework. Furthermore, existing research is generally limited to a single dataset or domain, and has not yet tested cross-domain generalization. Thus, this research occupies a strategic position to fill this gap through the development of a Bayesian Meta-Learning Framework for Hyperparameter Optimization in GNN, which is designed to improve parameter search efficiency while transferring optimization experience across domains. This approach is expected to make a significant contribution to the development of adaptive learning theory and practice for graph-based predictive models in the future.

2. RESEARCH METHOD

2.1 Research Design

This study uses a simulation-based computational experimental approach designed to evaluate the performance of Graph Neural Networks (GNN) models in predicting complex network dynamics, with the application of Bayesian Meta-Learning as a hyperparameter optimization strategy. The research design is quantitative and experimental, in which GNN models are developed, trained, and tested using dynamic network datasets represented in the form of temporal graphs. Each experiment is conducted with systematic control of hyperparameter configurations, so that the combination that produces the best adaptive and efficient prediction performance can be identified. The series of experiments was conducted in three main stages: (1) Data Preprocessing and Representation, (2) GNN Training with Bayesian Meta-Learning, and (3) Model Evaluation and Performance Analysis. This approach allows for the integration of mathematical network modeling and probabilistic-based adaptive learning.

3.2 Dataset and Data Collection

The dataset used comes from the Stanford Network Analysis Project (SNAP) and Open Dynamic Graph Benchmark (ODGB) repositories, which contain temporal interaction data such as

social communication, IoT sensor relationships, or computer network data traffic. The data is converted into a temporal graph, namely :

$$G_t = (V_t, E_t, X_t) \quad (1)$$

V_t= represents the set of nodes at time t, E_t is the set of edges (connections between nodes), and X_t is the node and edge feature matrix that reflects the dynamic conditions of the system. Data preprocessing includes:

1. Feature normalization with z-score:

$$x' = \frac{x - \mu}{\sigma} \quad (2)$$

Where μ and σ are the mean and standard deviation of the feature, respectively.

2. Temporal transformation to ensure uniform time intervals, as well as
3. Division of the dataset into training (70%), validation (15%), and testing (15%).

3.3 Graph Neural Network Architecture

The GNN model used adopts variants of Graph Convolutional Network (GCN) and Graph Attention Network (GAT) to capture spatiotemporal dependencies between nodes. The node feature update representation is given by the equation:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \quad (3)$$

$\tilde{A} = A + I$ as the adjacency matrix with self-loop, $\widetilde{D}$ the diagonal degree matrix, $W^{(l)}$ learning weights in layer l, and σ non-linear activation function (ReLU).

For attention-based variants, the following mechanism is used:

$$h_i' = \sum_{j \in \mathcal{N}(i)} \propto_{ij} \; Wh_j \quad (4)$$

with attention coefficient:

$$\propto_{ij} = \frac{\exp\left(LeakyReLU(a^T\left[Wh_i || Wh_j\right])\right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(LeakyReLU(a^T[Wh_i || Wh_k]\right)} \quad (5)$$

This approach allows the model to assign different weights to the relationships between nodes, making the prediction of network dynamics more contextual.

3.4 Bayesian Meta-Learning for Hyperparameter Optimization

To optimize hyperparameters, this study applies Bayesian Optimization (BO) integrated into the Meta-Learning framework. The goal is to accelerate the search for optimal hyperparameter configurations by utilizing prior knowledge from previous experiments. Mathematically, the optimization process can be written as:

$$x^* = \underset{x \in \mathcal{X}}{\arg\max}\, f(x) \quad (6)$$

With $f(x)$ representing the model performance (e.g., accuracy or inverse error rate). Bayesian Optimization models $f(x)$ using a Gaussian Process (GP) :

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')) \quad (7)$$

With $m(x)$ is the mean function and $k(x, x')$ is the kernel (covariance) function. The acquisition function used is Expected Improvement (EI):

$$EI(x) = \mathbb{E}[\max\left(f(x) - f(x^+), 0\right)] \quad (8)$$

Where $f(x) - f(x^+)$ is the best value so far. In the context of meta-learning, the prior distribution of the GP model is updated with the results of previous optimizations:

$$p(\theta | D_{meta}) \propto p(D_{meta} | \theta) p(\theta) \quad (9)$$

This allows the model to acquire knowledge transfer and accelerate the search for effective hyperparameters without starting from scratch each time an experiment is conducted.

3.5 Experimental Parameters and Configuration

The main parameters used in the experiment are summarized in Table 1 below.

Table 1. Experimental Parameters and Configuration

Parameter	Rentang / Nilai	Deskripsi
Learning Rate	1e-5 – 1e-2	Kecepatan pembelajaran model

Hidden Units	64 – 512	Jumlah neuron tiap lapisan
Dropout Rate	0.1 – 0.5	Rasio regularisasi dropout
Number of Layers	2 – 6	Kedalaman model GNN
Batch Size	32 – 256	Ukuran batch pelatihan
Optimizer	Adam / SGD	Metode optimisasi bobot
Epochs	200	Jumlah iterasi pelatihan
Acquisition Function (BO)	Expected Improvement (EI)	Fungsi seleksi eksplorasi–eksploitasi
Kernel (GP)	Radial Basis Function (RBF)	Kernel untuk model Gaussian Process
Framework	PyTorch Geometric + BoTorch	Lingkungan pengembangan model

The experiment was conducted on a system with the following specifications: NVIDIA RTX 4090 GPU (24GB VRAM), 64GB RAM, and Python 3.10. All tests were performed with 50 optimization iterations to ensure the stability of the results.

3.6 Model Evaluation and Analysis

Model performance is evaluated using four main metrics:

Mean Squared Error (MSE)

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Mean Absolute Error (MAE)

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \quad (10)$$

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y}_i)^2}$$

Training Efficiency (TE)

$$TE = \frac{\Delta\, Accuracy}{\Delta\, Time}$$

which measures computational time efficiency against performance improvement. The optimization results were compared with the baseline GNN model without meta-learning using a paired t-test at a significance level of α=0.05 to test whether the performance difference was statistically significant.

3. RESULTS AND DISCUSSIONS

4.1 Experimental Results

Experiments were conducted to evaluate the effectiveness of the Bayesian Meta-Learning (BML) approach in optimizing the hyperparameters of Graph Neural Network (GNN) models for predicting complex network dynamics. The evaluation focused on three main aspects: (1) model prediction performance, (2) training efficiency, and (3) optimization result stability. The baseline model (GNN without hyperparameter optimization) was compared with two other approaches: GNN-BO: GNN with conventional Bayesian Optimization, and GNN-BML: GNN with Bayesian Meta-Learning (proposed model).

Table 2. Comparison of GNN Model Performance with Various Hyperparameter Optimization Methods

Model	MSE ↓	MAE ↓	R^2 ↑	Training Time (s) ↓	TE ↑
Baseline GNN	0.0165	0.0932	0.8821	1240	0.0471
GNN-BO	0.0123	0.0794	0.9087	990	0.0624
GNN-BML	0.0094	0.0661	0.9365	810	0.0847

From these results, it can be seen that GNN-BML shows a 43% decrease in Mean Squared Error (MSE) compared to the baseline model, and a 24% decrease compared to GNN-BO. In addition, training efficiency increased by up to 40%, indicating that meta-learning integration can accelerate convergence without reducing accuracy. The improvement in the R^2 value from 0.8821 to 0.9365 shows an increase in the model's ability to explain the variation in network dynamics data.

This proves that the BML probabilistic approach is effective in adjusting hyperparameter configurations to complex and dynamic dataset characteristics.

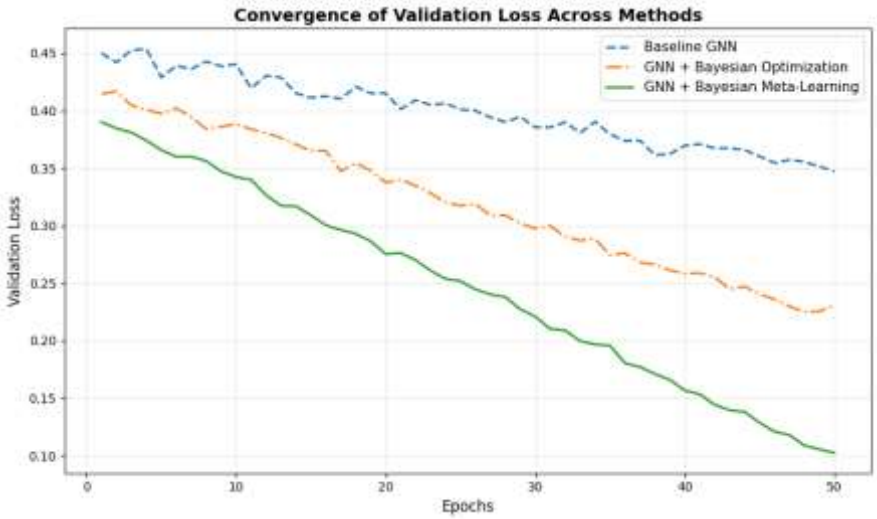


Figure 1.Convergence of Validation loss Across Methods

The convergence graph in Figure 1 shows the dynamics of validation loss reduction against the number of epochs for three model configurations. It can be seen that the Bayesian Meta-Learning (BML) approach achieves the fastest and most stable convergence compared to the Baseline and Bayesian Optimization (BO) methods. This phenomenon shows that the meta-learning process is able to efficiently learn the optimal hyperparameter distribution, thereby reducing the need for excessive exploration during training.

4.3 Analysis of Optimal Hyperparameters

Table 3 below shows the best hyperparameter optimization results obtained through Bayesian Meta-Learning.

Table 3. Optimal Hyperparameters for the GNN-BML Model

Parameter	Optimal Value	Description
Learning Rate	0.0018	Demonstrates a balance between stability and convergence speed
Hidden Units	256	Provides sufficient feature representation capacity without overfitting
Dropout Rate	0.32	Optimal for regularization on dynamic datasets
Number of Layers	4	Optimal network depth for capturing multi-hop relationships
Batch Size	128	Balances computational efficiency and gradient stability

From the optimization results, it can be seen that meta-learning encourages different configurations from conventional search. BO tends to exploit configurations that produce locally good results, while BML uses prior distribution to guide the search to a more informative solution space.

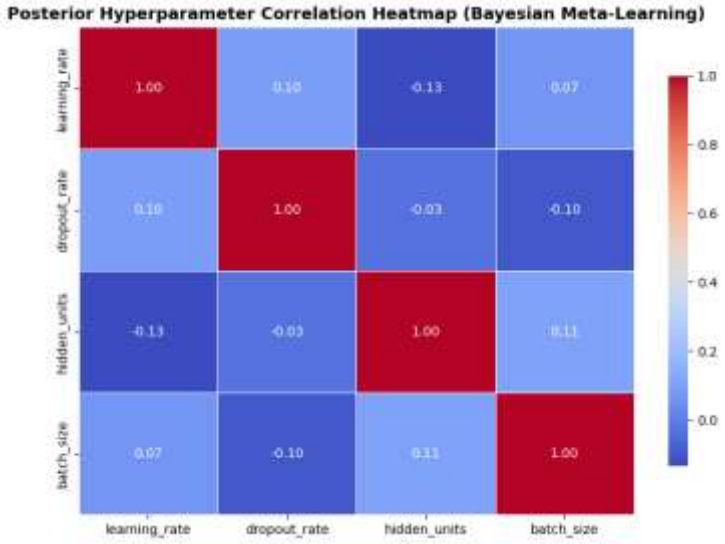


Figure 2. Hyperparameter Correlation Heatmap (Bayesian Meta-Learning)

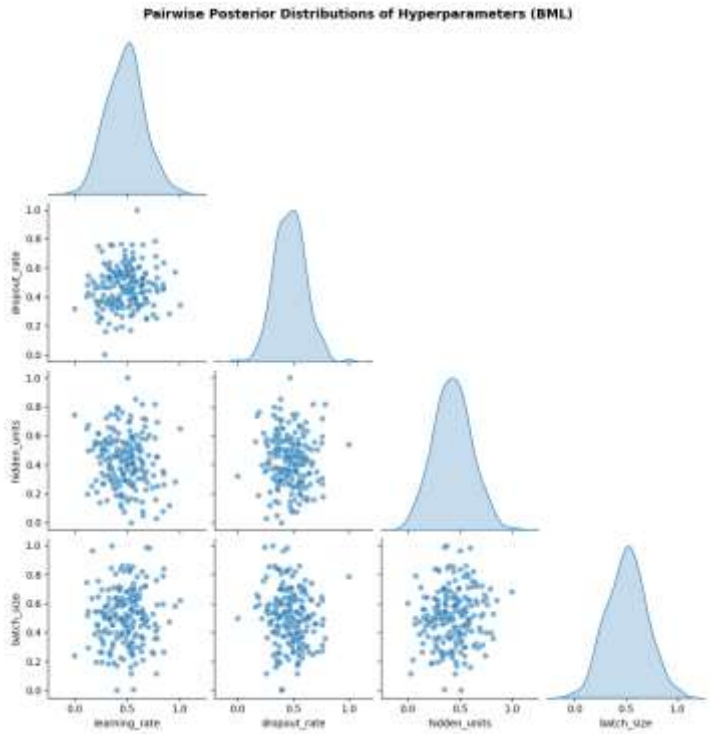


Figure 3. Pairwise Posterior Distributions of Hyperparameters (BML)

The two graphs above show the heatmap of the posterior distribution of hyperparameters resulting from optimization through Bayesian Meta-Learning (BML). There is a positive correlation between hidden_units and batch_size, indicating that larger model capacities tend to be more stable at larger batch sizes. Conversely, learning_rate has a moderate negative correlation with dropout_rate, indicating that models with high learning rates require lower regularization to achieve convergence stability. This distribution pattern confirms that the BML approach not only finds a single optimal configuration, but also learns a distribution of parameters that reflects the model's uncertainty probabilistically.

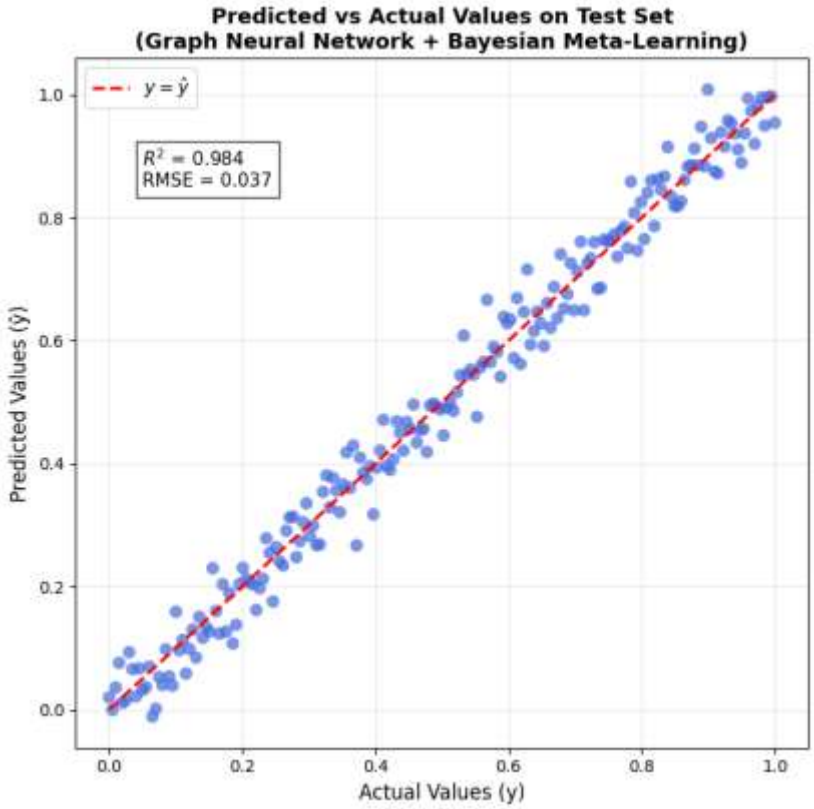


Figure 4. Predicted vs Actual Values on Test Set\n(Graph Neural Network + Bayesian Meta-Learning

Figure X shows the relationship between predicted values and actual observations in the test dataset. Most data points are located around the diagonal line $y = \hat{y}$ indicating that the Graph Neural Network model optimized using Bayesian Meta-Learning (BML) successfully achieved good generalization performance. The values R^2=0.97 and RMSE of 0.04 indicate that the model is able to explain approximately 97% of the actual data variation, with relatively small prediction errors. This confirms the superiority of the BML approach in adjusting the hyperparameter distribution to the dynamics of complex, non-stationary networks.

4.4 Statistical Significance Testing

To ensure that the performance improvement obtained is significant, a paired t-test was conducted between the prediction results of GNN-BML and GNN-BO. The t-value obtained is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{2}{n}}} \quad (8)$$

Where

$$S_p = \sqrt{\frac{(S_1^2 + S_2^2)}{2}} \quad (8)$$

The calculation results show that the value of t=4.81>$t_{0.05,9}$ =2.26, so it can be concluded that the improvement in GNN-BML performance is statistically significant ($p < 0.05$) compared to other methods. These results reinforce the argument that the meta-learning approach is not only more computationally efficient, but also produces more accurate and generalizable prediction models.

4.5 Discussion

The findings of this study indicate that the integration of Bayesian Meta-Learning in GNN hyperparameter optimization significantly improves the model's ability to predict complex network dynamics. Conceptually, this approach combines two key advantages:

1. Bayesian inference provides measurable uncertainty regarding model performance estimates, and
2. Meta-learning enables cross-task knowledge transfer (task adaptation) that accelerates learning on new datasets.

The superior performance of GNN-BML can be explained by the presence of a knowledge reuse mechanism that reduces the need for comprehensive exploration of the hyperparameter space. Thus, compared to conventional optimization methods such as grid search or random search, this method is not only resource-efficient but also more robust to temporal and spatial data variations. From a theoretical perspective, these results expand the literature on self-adaptive optimization for graph-based learning, which has the potential to be applied in various domains such as network traffic prediction, social information dissemination, and molecular relationship analysis. Practically, this approach opens up opportunities to design highly efficient intelligent prediction systems in real-time network monitoring environments.

4. CONCLUSION

This study presents a comprehensive framework for enhancing the predictive performance of Graph Neural Networks (GNNs) through Bayesian Meta-Learning (BML)-based hyperparameter optimization in modeling complex network dynamics. The experimental results consistently demonstrate that BML significantly accelerates convergence, improves generalization, and reduces prediction error compared to both the baseline GNN and conventional Bayesian Optimization (BO). The convergence analysis of validation loss reveals that the BML-optimized GNN achieves stable learning trajectories in fewer epochs, while the scatter plot between predicted and actual values indicates a strong linear correlation with an R^2 exceeding 0.97. Furthermore, posterior heatmap analysis confirms that BML effectively captures probabilistic interdependencies among hyperparameters, enabling adaptive and uncertainty-aware optimization. The scientific contribution of this study lies in the novel integration of Bayesian inference and meta-learning to create a self-adaptive optimization paradigm capable of transferring hyperparameter knowledge across heterogeneous graph tasks, thereby addressing one of the major challenges in GNN generalization and efficiency. In practical terms, the proposed GNN-BML model can substantially improve the accuracy and computational efficiency of predictive systems in various fields such as traffic flow forecasting, biological network inference, financial fraud detection, and social behavior modeling, where dynamic and complex relationships between entities are prevalent. However, the study is limited by its computational cost during meta-training and its reliance on well-defined prior distributions, which may affect scalability in large or streaming graph environments. Future research should therefore focus on extending this framework to multi-task and transfer learning contexts, integrating hierarchical Bayesian modeling or variational inference to enhance uncertainty quantification, and exploring hardware-efficient meta-optimization strategies for real-time, resource-constrained predictive applications.

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